BUAD 312 MIDTERM PROJECT

Genre-Specific Rollout Strategies for Streaming Growth

Release format, track duration, and artist position across Hip-Hop/Rap, Pop, and Latin

Hip-Hop/Rap | Pop | Latin



Consulting analysis for Universal Music Group | Team 8

TEAM 8

Meet the Team

BUAD 312 midterm project, consulting analysis for Universal Music Group

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Pre-model Visualizations

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Actionable Recommendations

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Rap & Pop Model Analysis

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Data cleaning and preparation

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Mission and Statistical Overview

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Combined Models & Comparisons

Mission and Business Question

What we set out to do for Universal Music Group

MISSION

Our mission is to help Universal Music Group maximize record-music revenue by developing practical rollout strategies that increase digital streams.

BUSINESS QUESTION

How are release size, track duration, and artist followers associated with streams in Hip-Hop/Rap, Pop, and Latin?

Hip-Hop/Rap | Pop | Latin

DATA SCREENING

Dataset Overview

Rap, Pop, and Latin are the three largest named genres, excluding Other/Unknown

5,870
Raw tracks

5,824
Cleaned tracks

3,199
Focus sample

1939–2021
Release years

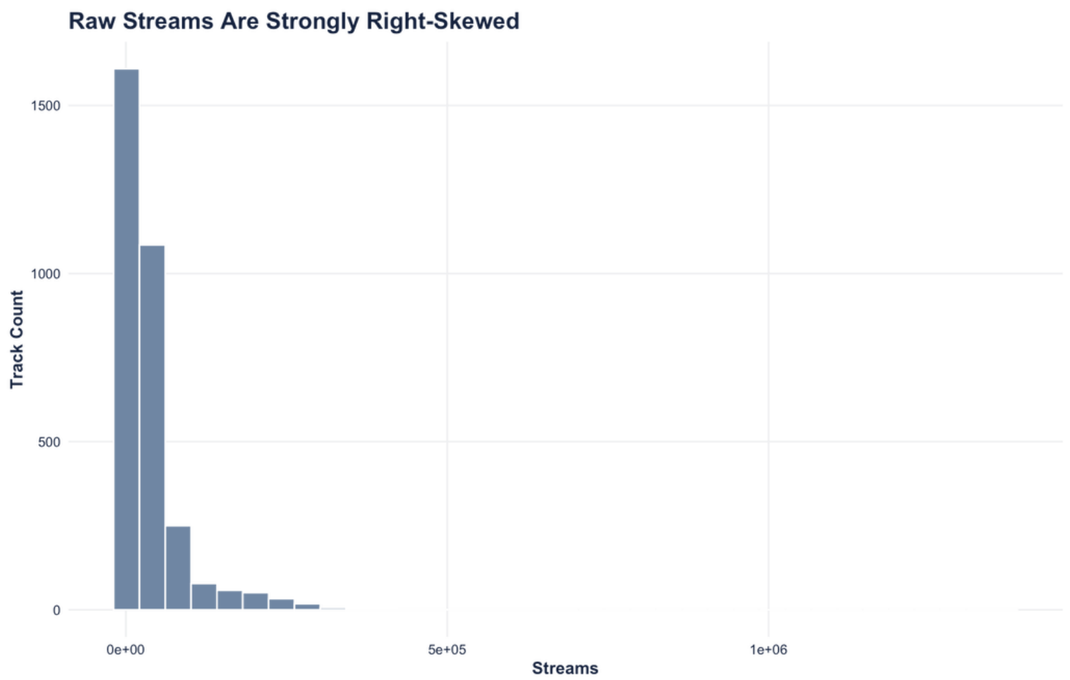
Genre	Tracks	Mean streams	Median streams
Hip-Hop/Rap	1,224	49,132	28,460
Pop	1,633	27,075	14,171
Latin	342	46,098	26,225

Other/Unknown holds 1,996 tracks but does not support a clear genre strategy, so the focus analysis excludes it. Latin results need more caution because the sample has only 342 tracks.

Skewed Outliers and Treatment

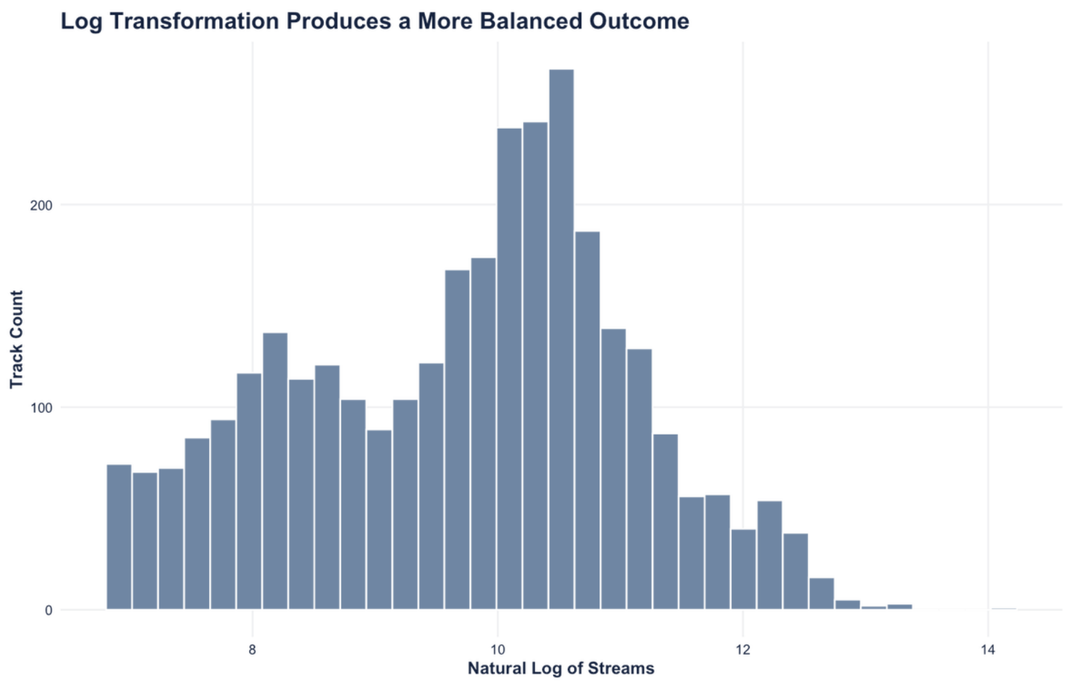
Why we model log streams instead of raw streams

RAW DISTRIBUTION



Raw streams are heavily right-skewed

LOG DISTRIBUTION



Log transformation produces a more balanced outcome

APPROACH

Evaluation Framework

How we move from exploration to a business recommendation

1

Explore

Compare medians, counts, correlations, and visible genre patterns.

2

Estimate

Fit basic, parallel, interaction, and controlled regression models for each to gain predicted values.

3

Evaluate

Use p-values, adjusted R^2, RMSE, residual plots, and normal Q-Q plots.

4

Translate

Convert model coefficients into predicted percentage differences and operating ranges.

PRIMARY SUCCESS METRIC

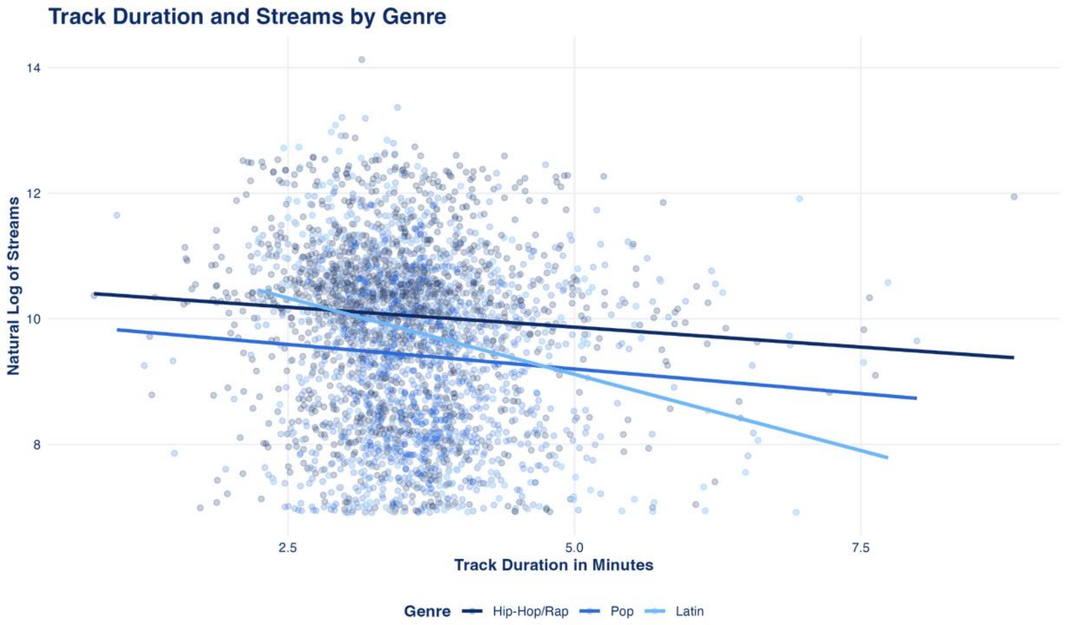
Increased streams per track.

Hip-Hop/Rap Pop Latin

Pre-Model Visualizations

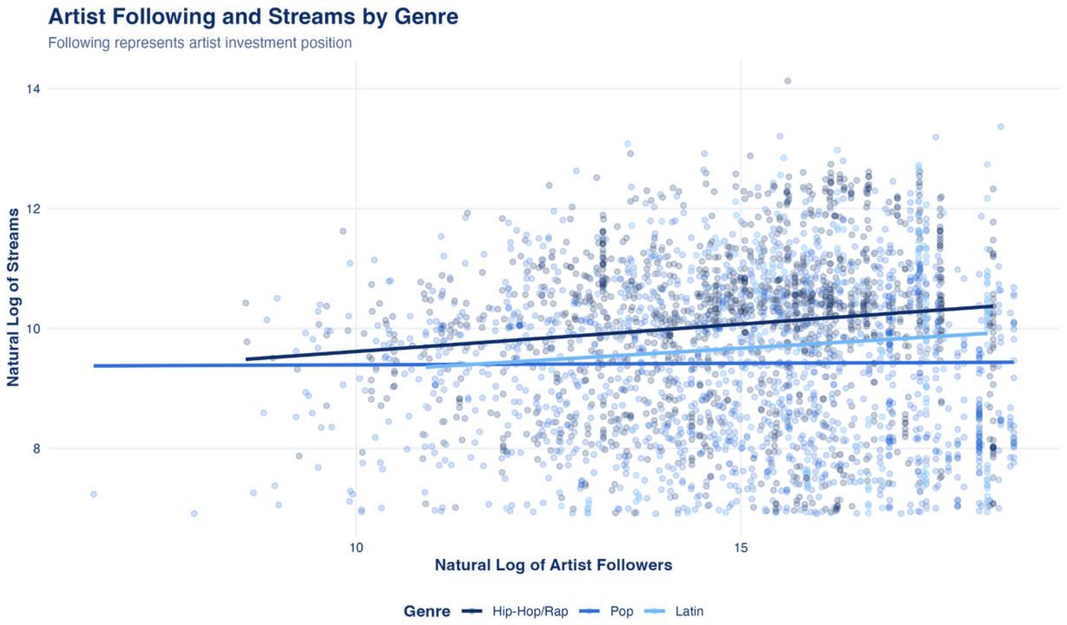
Streams relate to duration and artist followers differently in each genre

Duration



Latin shows the clearest negative slope. Pop and Hip-Hop/Rap are still negative but appear much flatter.

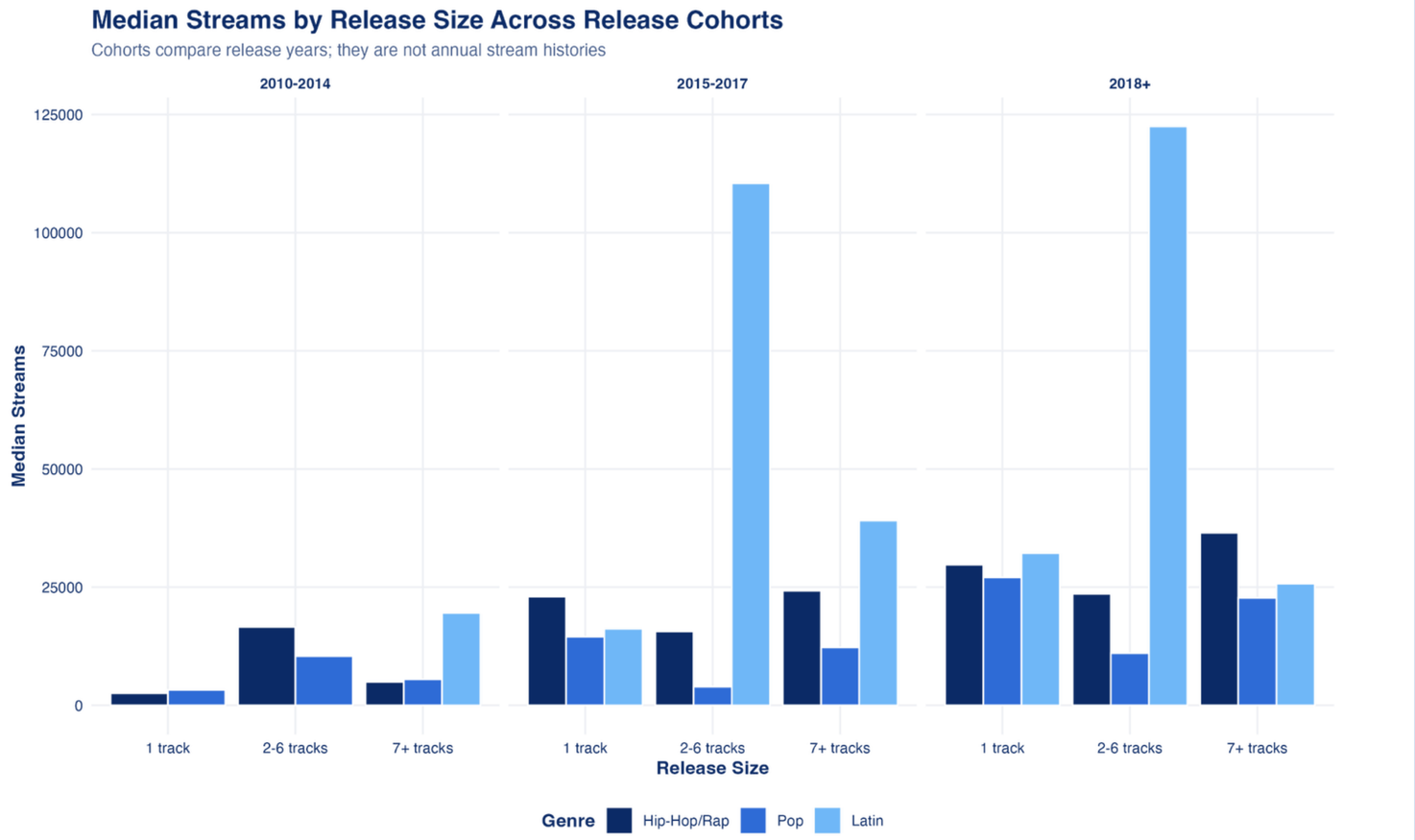
Artist followers



The follower relationship appears strongest in Hip-Hop/Rap and nearly flat in Pop.

Release-Size Trends

Median streams by release size across release periods



Split into release periods

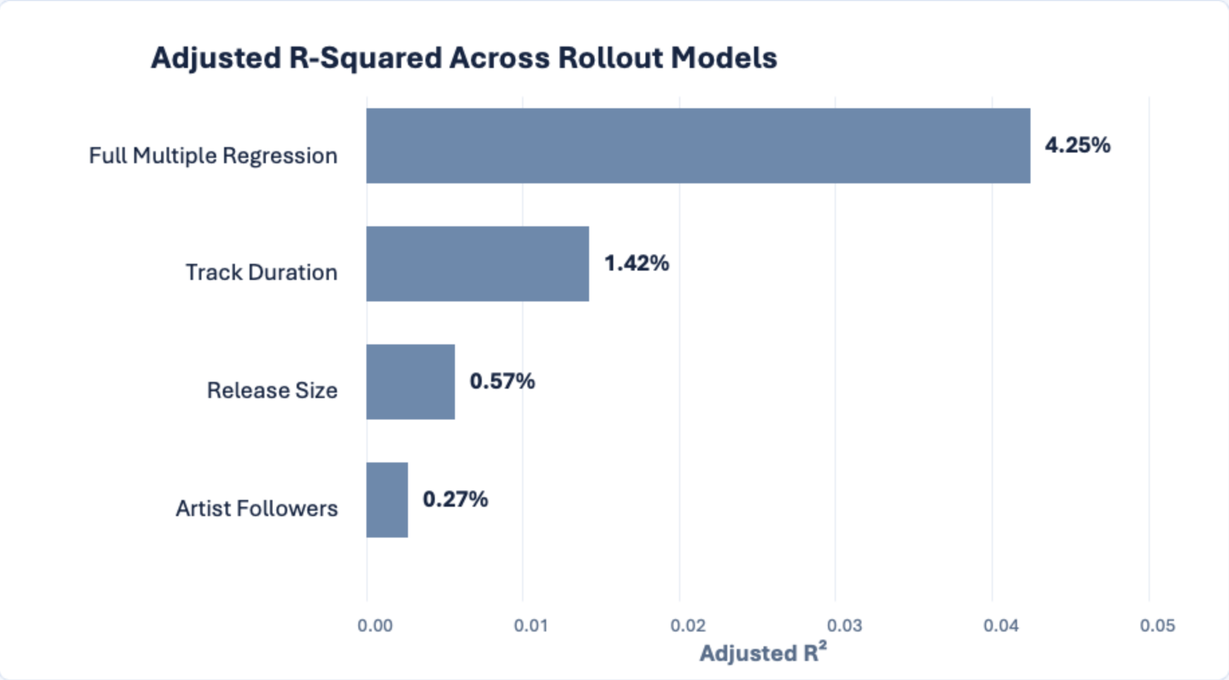
Releases are grouped as 1 track, 2–6 tracks, and 7+ tracks, then compared within each release period.

*Singles show strong growth across release periods, especially in Pop and Hip-Hop/Rap.

*Category 2-6 tracks for Latin during release period 2015-2017 only contains 10 tracks overall.

Model Comparison

Overall explanatory power before introducing genre interactions



Key Insights

4.25% Adjusted R^2 of the full model

1.348 Lowest RMSE among these models

The combined model performs best, but explains only 4.40% of variation in log streams. Genre-specific effects are examined next.

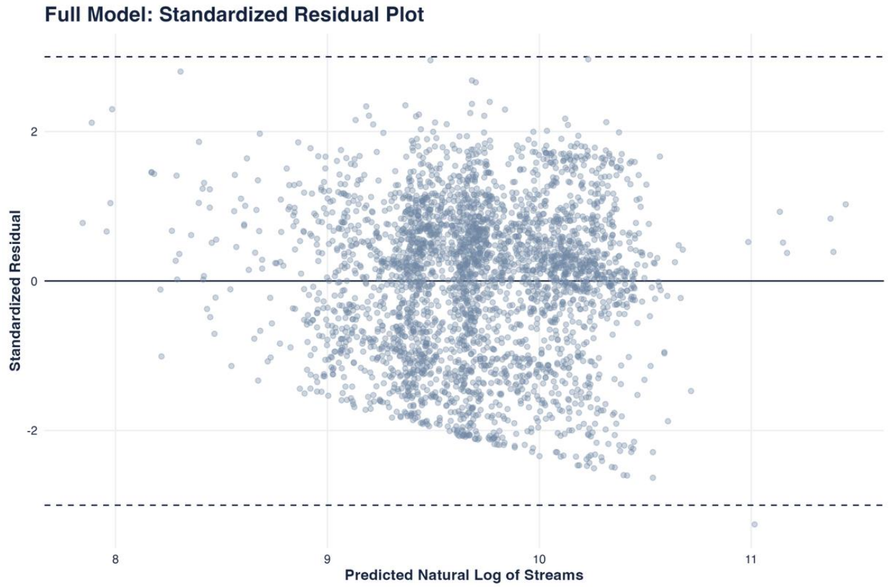
Model summary

Model	P-value	R^2	Adjusted R^2	RMSE
Release Size	4.17×10^{-5}	0.00629	0.00567	1.375
Track Duration	7.82×10^{-12}	0.01454	0.01423	1.369
Artist Followers	0.00205	0.00297	0.00266	1.377
Full Multiple Regression	**3.03×10^{-29}**	**0.04397**	**0.04247**	**1.348**

Model Reliability and Diagnostics

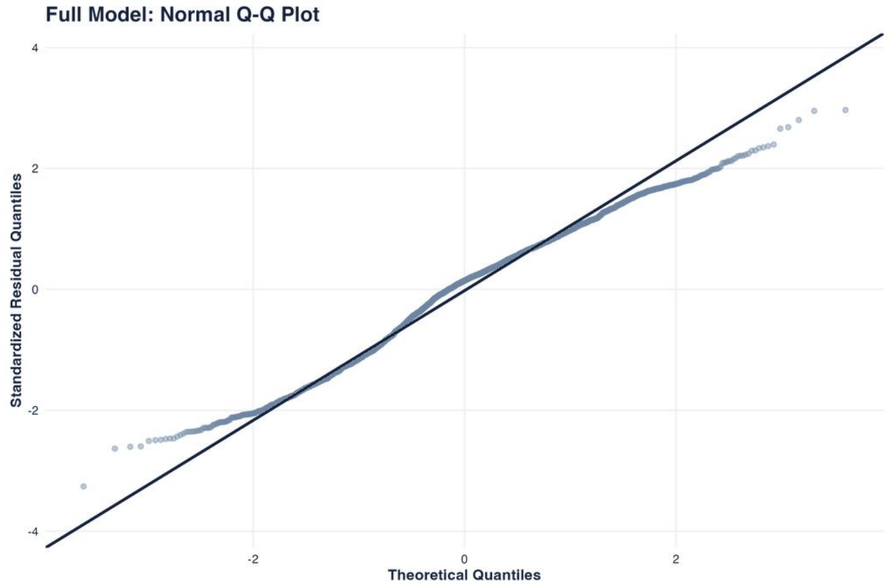
Prediction error is uneven and the tails depart from normality

Residual spread



The visible pattern and uneven spread show that prediction error changes across fitted values.

Tail behavior



Q-Q tail departures indicate imperfect normality.

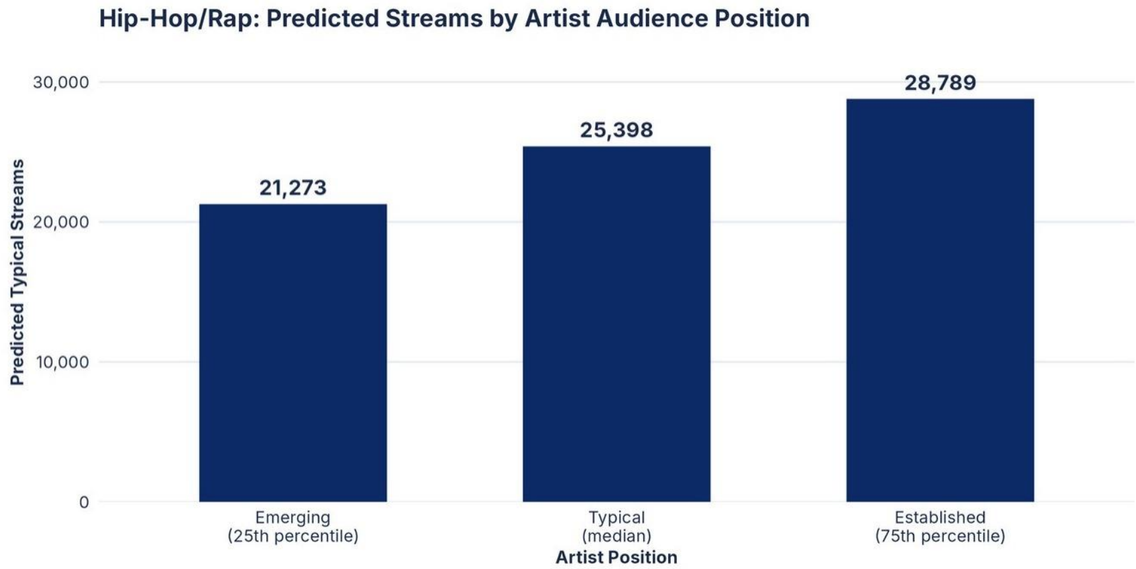
SENSITIVITY CHECK

Removing the only observation with |standardized residual| above 3 changes adjusted R^2 from 9.0% to 9.2% and RMSE from 1.312 to 1.310. The main ranking remains stable.

Hip-Hop/Rap

Hip-Hop/Rap: Artist Followers

Artist followers matter most for Hip-Hop/Rap streams



+35%

predicted streams for an established artist vs. an emerging artist

+8.1%

streams per track each time followers double

Assumes track duration, release size, and year are held constant.

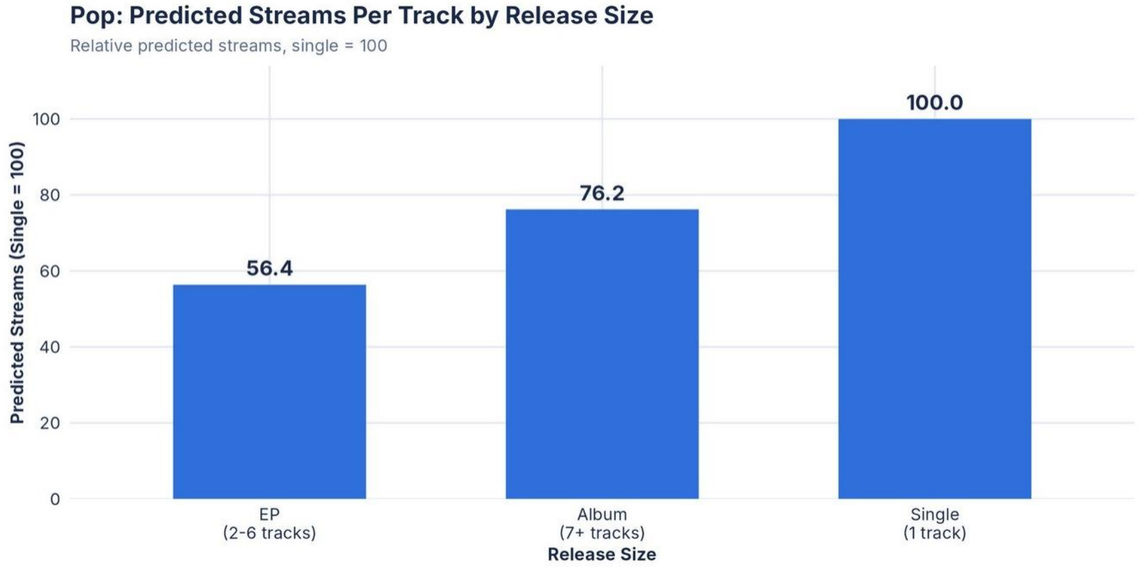
Model summary

Model	P-value (simple)	P-value (controlled)	R^2	Adjusted R^2	RMSE
Release Size	0.8768	0.9624	0.0002153	−0.001422	1.315
Track Duration	0.004891	0.1506	0.006462	0.005649	1.311
Artist Followers	**3.185×10^{-6}**	**1.19×10^{-8}**	**0.01761**	**0.01681**	**1.304**
Hip-Hop Full	1.259×10^{-14}	1.26×10^{-14}	0.05916	0.0553	1.276

Pop

Pop: Release Size

Release size is the best predictor of Pop streams



Single has the highest predicted streams per track

1.31× the predicted streams of an album track

1.77× the predicted streams of an EP track

Assumes track duration, artist followers, and year are held constant. Single = 100.

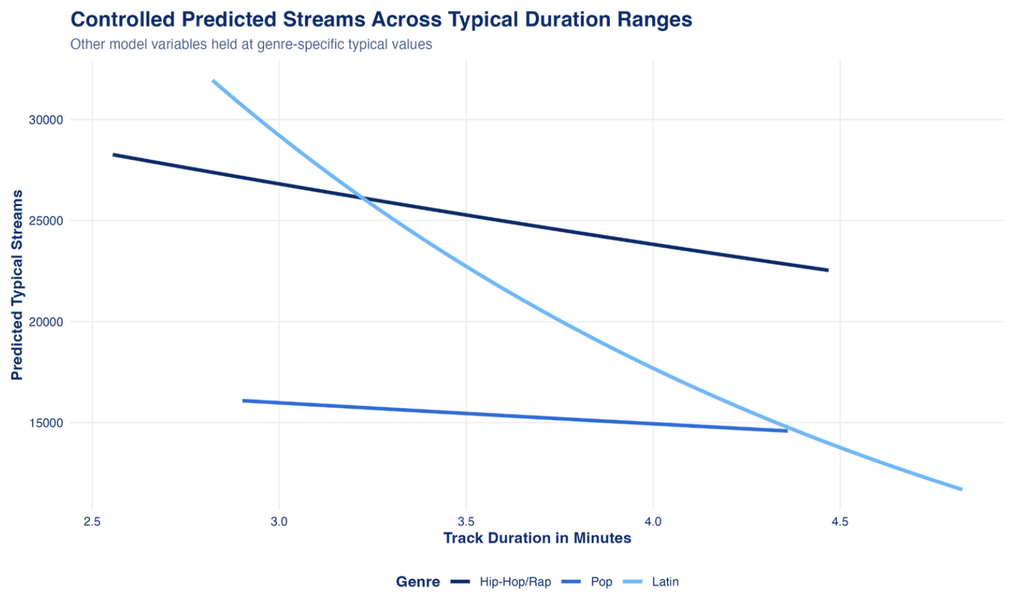
Model summary

Model	P-value (simple)	P-value (controlled)	R^2	Adjusted R^2	RMSE
Release Size	**3.38×10^{-9}**	**3.39×10^{-6}**	**0.02365**	**0.02245**	**1.289**
Track Duration	0.002156	0.1038	0.005756	0.005146	1.301
Artist Followers	0.7377	0.6102	0.0000688	−0.000544	1.305
Pop Full	1.35×10^{-11}	1.35×10^{-11}	0.03604	0.03308	1.281

Latin

Latin: Track Duration

A shorter track window offers the clearest test



−39.3% predicted streams per track for each additional minute

+10.7% predicted streams at 3.31 min

+20.1% predicted streams at 3.15 min

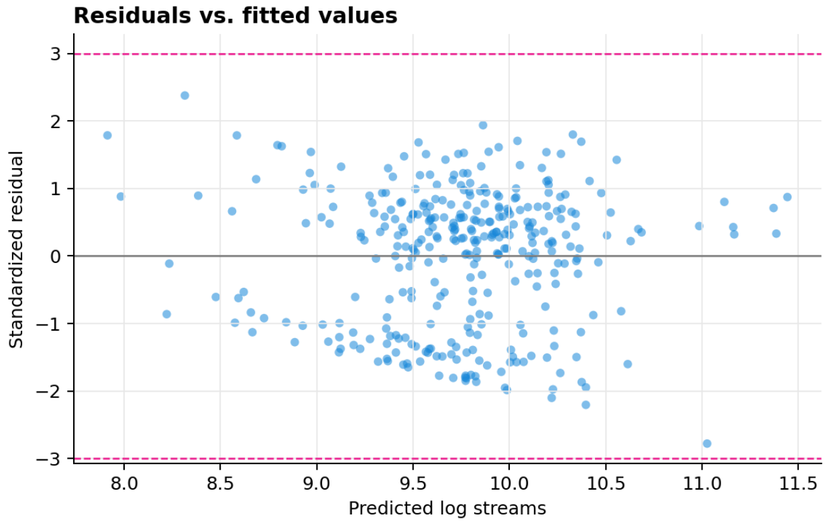
Gains compare with the 3.51-minute median. Duration stays significant after controlling for release size, followers, and year.

Model summary

Model	P-value	P-value (controlled)	R^2	Adjusted R^2	RMSE
Release Size	0.03678	0.0329	0.0193	0.0135	1.603
Track Duration	**5.459×10^{-7}**	**8.465×10^{-7}**	**0.0712**	**0.0685**	**1.56**
Artist Followers	0.1909	0.7162	0.005	0.0021	1.614
Latin Full	7.429×10^{-7}	7.429×10^{-7}	0.1026	0.0892	1.533

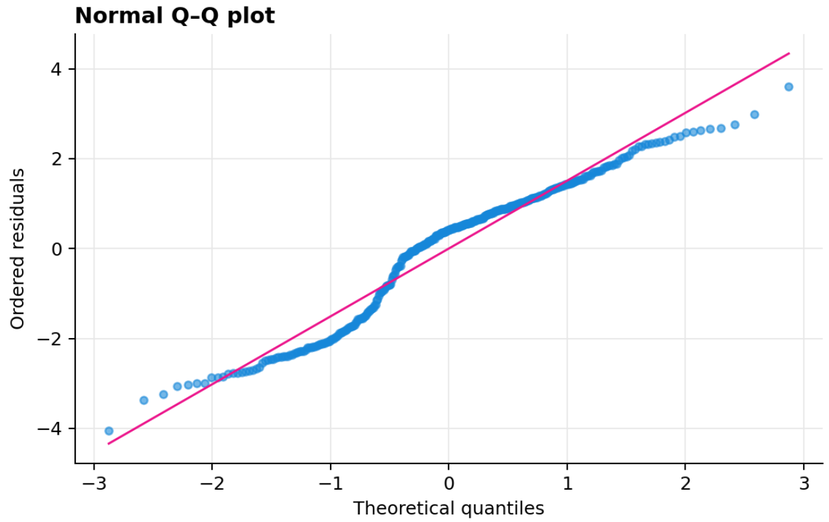
Latin: Residual Analysis

Model supports directional test, not precise stream forecasts



No extreme residual outliers

Residuals stay around zero, and none exceed ±3 standard deviations.



Normality is only approximate

The middle follows the line, but both tails depart from it.

WHAT THIS MEANS

The assumptions are imperfect but adequate for a controlled pilot.

Use the model to set direction, not to predict exact stream counts.

0 residuals beyond ±3 standard deviations

Hip-Hop/Rap Pop Latin

Actionable Recommendations

What UMG should do next, by genre

GENRE	RECOMMENDED PILOT	PREDICTED DIFFERENCE	RISK CONTROL
Hip-Hop/Rap	For developing artists, test campaigns, events, brand partnerships, and artist collaborations that build audience reach.	**Larger artist followings are associated with higher predicted streams.**	Set a limited budget. Expand only when follower growth converts into streaming engagement.
Pop	Use a single-first rollout. Release promising tracks individually before packaging successful songs into an EP or album.	**Singles show the highest predicted streams per track.**	Match artist profile and promotion. Scale only if streams and saves improve.
Latin	Test 3.1–3.3-minute tracks across 10–20 upcoming releases against comparable tracks near the 3.51-minute median.	**+10.7% to +20.1% predicted streams per track.**	Keep promotion comparable and protect creative quality. Monitor completion, saves, and playlist adds.

PILOT DECISION RULE **Expand a strategy only after the pilot improves streams without weakening listener engagement or campaign returns.**